

GROUP 14 | CLIMATE & AGRICULTURE CHALLENGE

Eastern Syria Agro-Climate

Assessing Water Stress & Drought Risks on Rainfed Agriculture

Focus Area: Eastern Syria (Al-Jazira Region) • Tools: Python, NASA POWER& SPI Index

PROJECT TEAM MEMBERS (GROUP 14)

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01 / Problem Statement

The Climate Crisis in Al-Jazira

- Eastern Syria (Al-Jazira) is the nation's primary agricultural breadbasket.
- Intensifying climate change has brought erratic rainfall patterns, soaring temperatures, and frequent droughts.
- Strategic rainfed crops like wheat and barley are facing severe, recurring yield declines.

Current Gap & Proposed Solution

- Operational Gap: Stakeholders currently rely on traditional methods lacking historical analytics & intelligent forecasting.
- Our Approach: Utilizing Python to process 20 years of historical meteorological metrics & Standardized Precipitation Index (SPI).
- Goal: Pinpoint high-vulnerability zones and critical drought periods to support early response & enhance food security.

02 / Research Question & Datasets

Main Research Question

“How have shifting rainfall patterns and extreme temperatures impacted agricultural water stress indicators and drought frequency in Eastern Syria over the past two decades?”

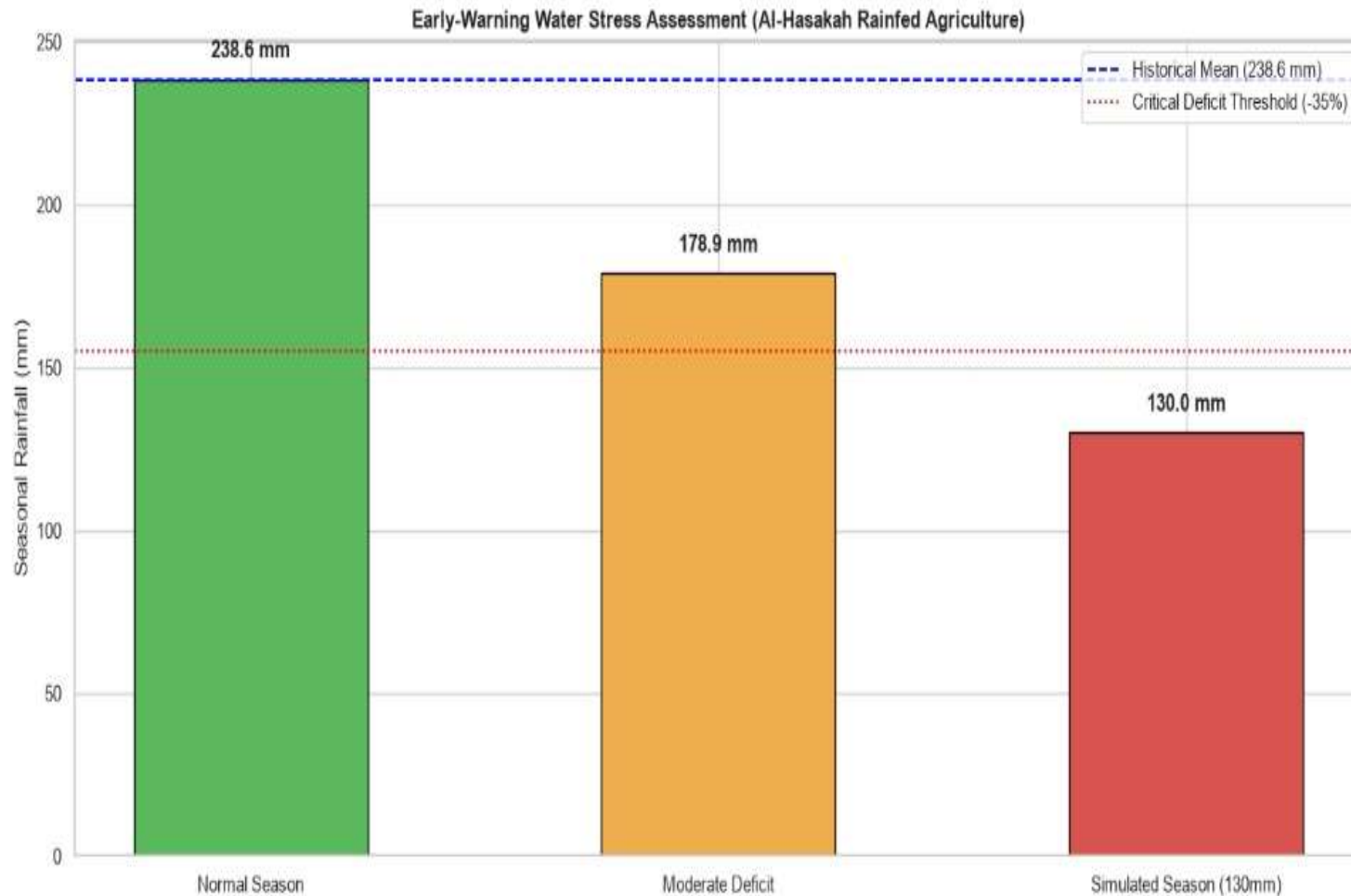
This inquiry drives our quantitative analysis to connect long-term climate trends directly with real-world agricultural impact.

Data Sources & Telemetry

NASA POWER Data Portal

- Time-series meteorological data: precipitation, temperatures, and solar radiation.
- URL: power.larc.nasa.gov

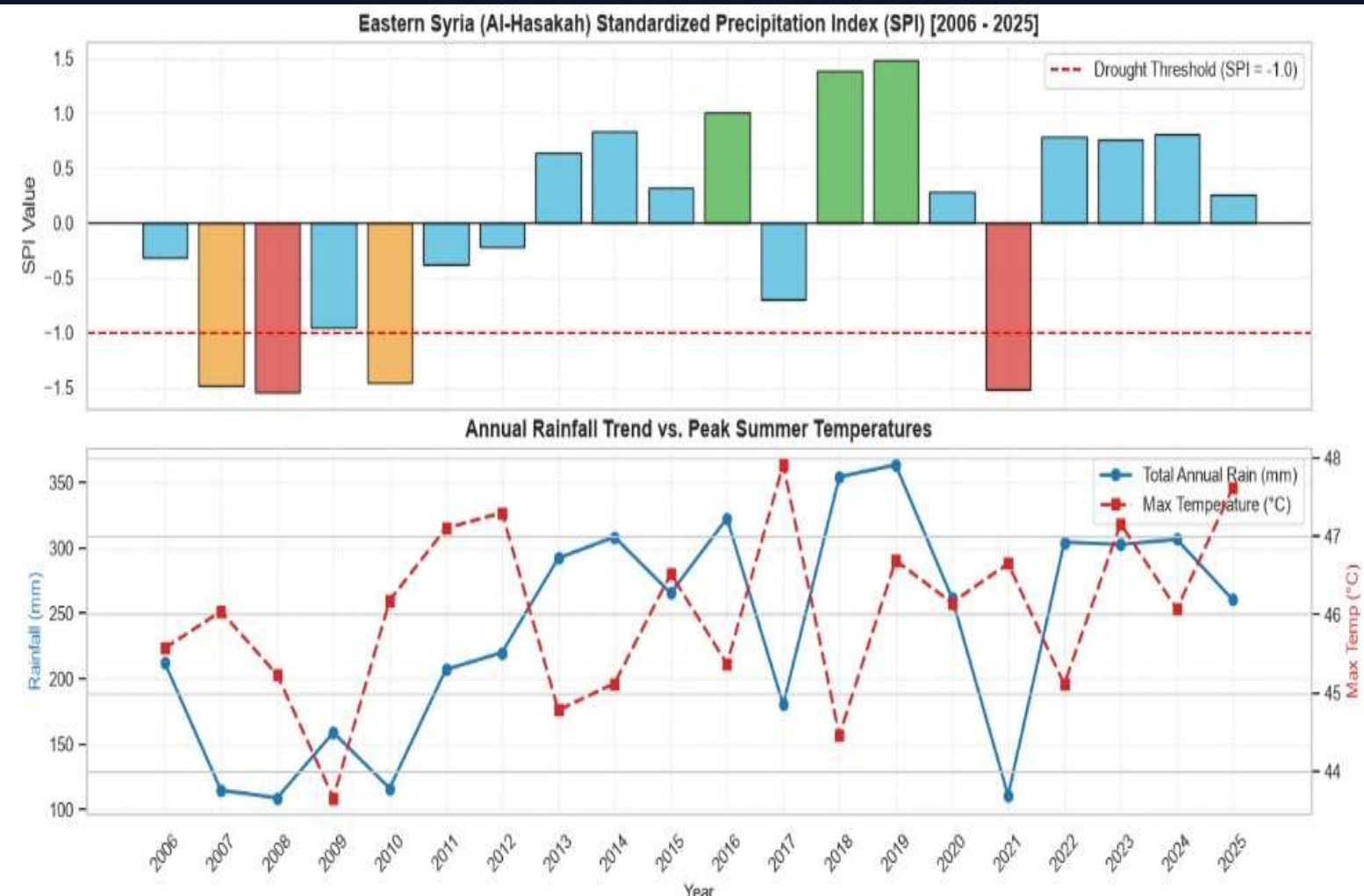
03 / Climate Trends & SPI Analysis (2006 - 2025)



Key Analytical Insights

- Severe Drought Events: SPI drops well below the critical threshold (-1.0) in years 2007, 2008, 2010, and 2021.
- Inverse Correlation: Prolonged rainfall deficits directly coincide with peak summer temperature spikes (approaching 48°C).
- High Vulnerability: The Al-Hasakah rainfed region experiences sharp multi-year precipitation swings, endangering crop growth.

04 / Early-Warning Water Stress Assessment



Water Stress Thresholds

- Historical Baseline: Normal average seasonal rainfall stands at 238.6 mm for rainfed agriculture.
- Critical Deficit Threshold: Set at -35% reduction (~155 mm), below which severe crop failure occurs.
- Simulated Crisis Scenario: Simulated drought season reaches 130.0 mm, crossing into high risk zone requiring early emergency response.

05 / Methodology & Expected Output

Proposed Methodology

► Data Cleaning & Processing

Python pipeline utilizing Pandas and NumPy for missing value handling, time-series alignment, and aggregation.

► SPI Calculation & Visualization

Computing the Standardized Precipitation Index (SPI) and plotting trends with Matplotlib & Seaborn.

► Drought Pattern Mapping

Spatial-temporal mapping to identify vulnerable agricultural zones across Al-Jazira.

Expected Output

- ✓ Identification of Critical Periods: Precise mapping of historical water-stress periods over 20 years.
- ✓ Prototype Early-Warning Framework: Data-driven decision tool for early agricultural intervention.
- ✓ Actionable Policy Insights: Guiding local farmers and NGOs towards resilient crop planning and water management.

o6 / Future Roadmap & Policy Recommendations

Policy Recommendations for Decision-Makers & NGOs

- Adaptive Crop Management: Shift towards drought-tolerant and heat-resistant wheat/barley varieties during predicted high-risk seasons.
- Supplementary Irrigation Infrastructure: Expand rainwater harvesting systems and localized groundwater recharge in vulnerable rainfed zones.
- Proactive Relief Aid: Trigger early emergency support and subsidized seed distribution when SPI forecast drops below -1.0 before the planting season.

Future Technical Roadmap

- Machine Learning Integration: Implement predictive LSTM / XGBoost models to forecast seasonal SPI 3 to 6 months in advance.
- GIS & Satellite Imagery: Integrate NDVI (Normalized Difference Vegetation Index) from Sentinel/Landsat for real-time crop health monitoring.